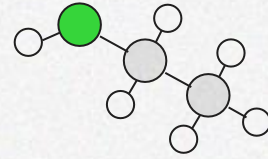


System Setup

2026 i-CoMSE Workshop

Kayla G. Sprenger





Overview

Session 1: How do I choose the right protein and evaluate it?

Session 2: How do I get the right ligand bound to that protein?

Session 3: How do I prepare the complex for simulation?

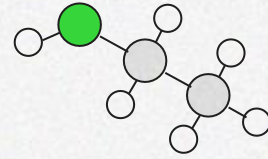




Session 1

Choosing and
Preparing the
Starting Protein
Structure

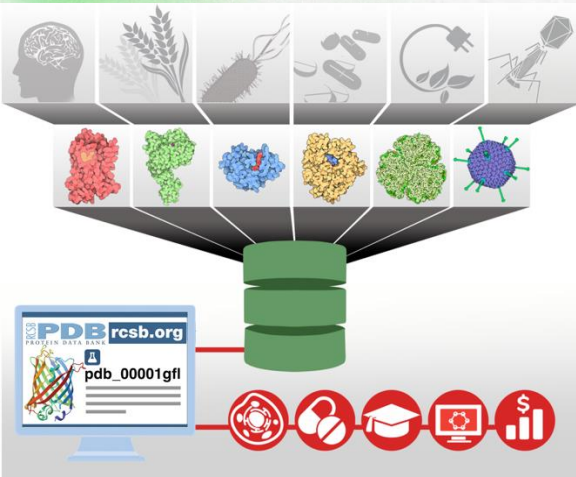
Learning Objectives



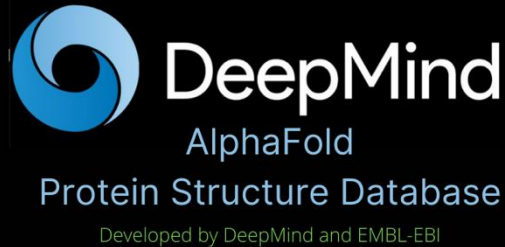
- **Find** a suitable starting protein structure (*Where do I get a structure?*)
- **Evaluate** whether a structure is suitable for simulation (*Can I trust it?*)
- Identify missing information in order to **Repair and Refine** the structure (*How do I fix or improve it?*)
- Decide when to use an experimental structure, AlphaFold prediction, or homology modeling (*What's the best starting point for my system?*)
- **Apply** this workflow to select and prepare a starting structure for molecular dynamics simulations



Find a Starting Protein Structure



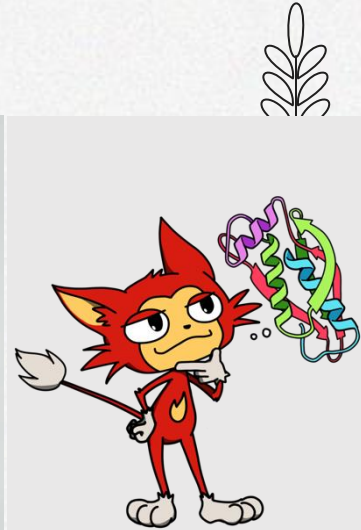
Use **RCSB PDB**
for experimental
coordinates

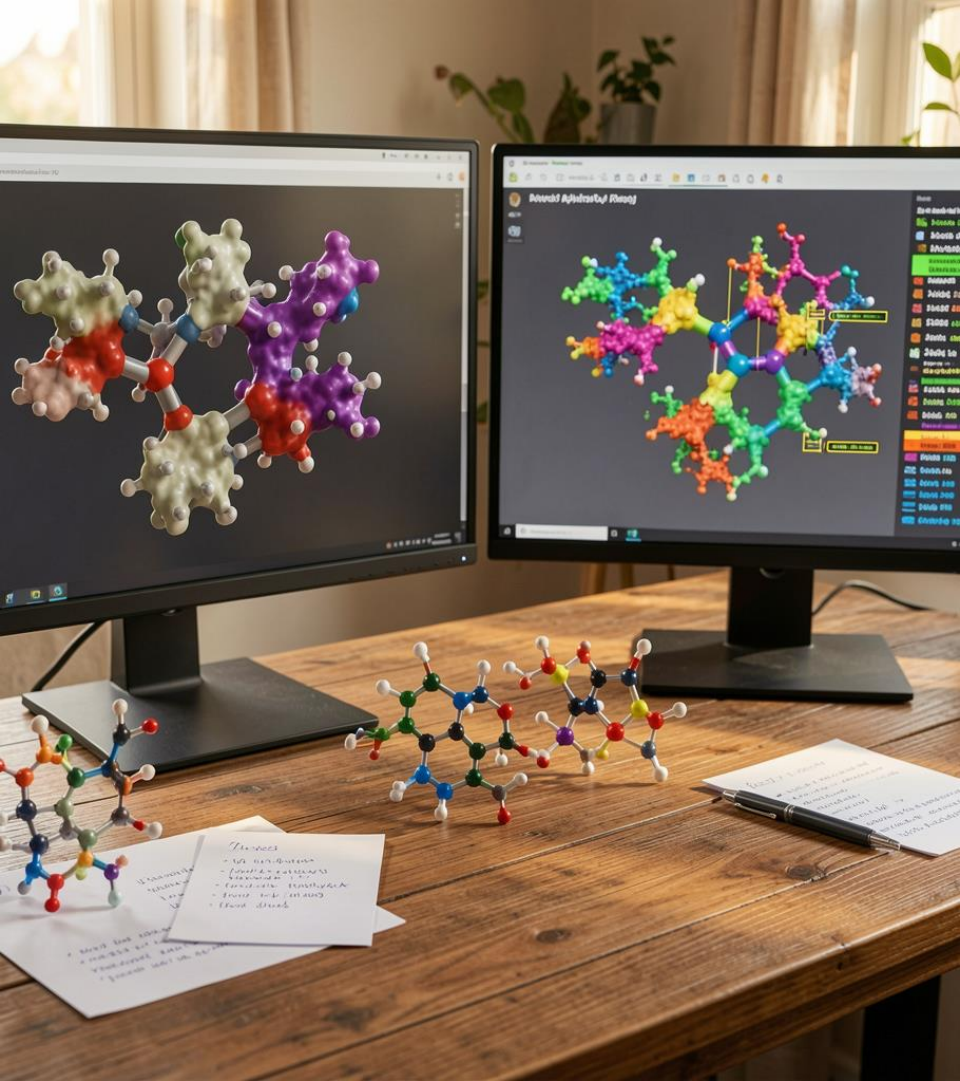


Use **AlphaFold DB**
for pre-computed
predicted models



Use **AlphaFold3** or **ColabFold**
for predicting new models





Evaluate the Starting Protein Structure: *Quality, Completeness, & Biological Relevance*

Live PDB entry walkthrough:
7AAA: PARP1 catalytic domain

Download 7AAA.pdb:

<https://www.rcsb.org/structure/7AAA>

Evaluate: *Quality*

- Highest resolution? **1.74 Å**
- X-ray, cryoEM, or NMR? **X-ray diffraction**
- Alternate residue conformations? **No**

- `awk '($1=="ATOM" || $1=="HETATM") && substr($0,17,1)!=" " {print}' 7AAA.pdb`

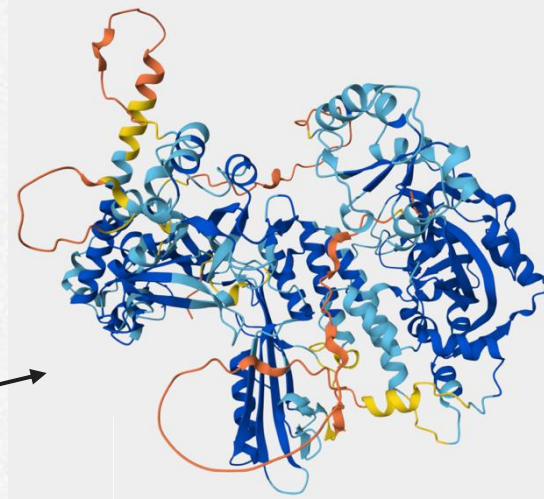
- Structural confidence? **Very high**

- Depends on Technique:

- For X-ray: Resolution →

- For AlphaFold:

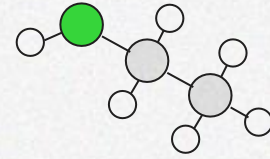
- Per-residue confidence score (pLDDT)



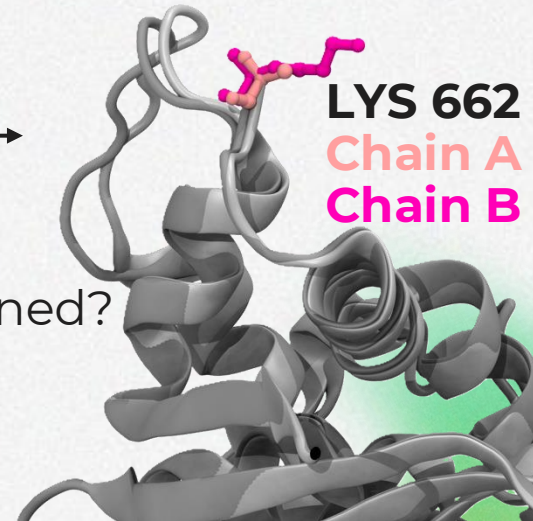
Resolution	Confidence
<1.5 Å	Outstanding
1.5–2.0 Å	Excellent
1.5–2.0 Å	Very good
2.0–2.5 Å	Moderate
2.5–3.0 Å	Low for detailed MD
3.0–3.5 Å	Use with caution

Terminal Command

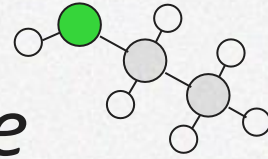
Evaluate: *Completeness*



- Missing domains or residues? **Yes (GLY 660 and SER 661)**
 - `grep -A 10 "MISSING RESIDUES" 7AAA.pdb`
 - If **Yes**, do these need to be modeled in? **No (Expression tags)**
 - `grep SEQADV 7AAA.pdb`
 - If **Yes**, use homology modeling (MODELLER)
- Missing atoms? **Yes (e.g., in LYS 662)** →
 - `grep -A 17 "MISSING ATOM" 7AAA.pdb`
 - If **Yes**, use PDBFixer
- Which (if any) crystallographic waters should be retained?
 - `grep HETATM 7AAA.pdb | grep HOH | wc -l` **None**
- Missing cofactors or biologically relevant ions? **No**
 - If **Yes**, use AlphaFill

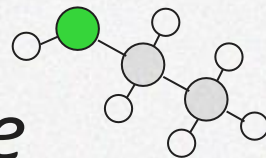


Evaluate: *Biological Relevance*



- Truncated domains? **Yes (only catalytic domain)**
 - If **Yes**, do we need to cap the termini? **Yes**
 - `grep '^ATOM' 7AAA.pdb | head -1`
 - `grep '^ATOM' 7AAA.pdb | tail -1`
 - Construct starts at 662, ends at 1011 → these become artificial termini. Add neutral 'caps' to N-term (ACE) and C-term (NME)
- Fusion or purification tags? **Yes**
 - `grep SEQADV 7AAA.pdb`
- Apo or holo? **Apo (no inhibitor)**
- Bound ligands? **No (only crystallization ligands – remove!)**
- Species? **Human (homo sapiens)**

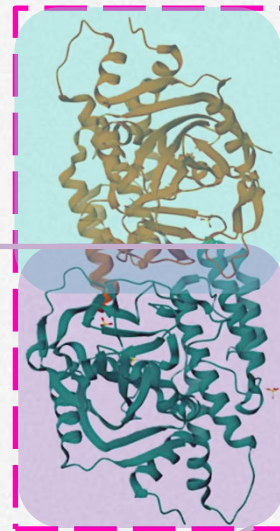
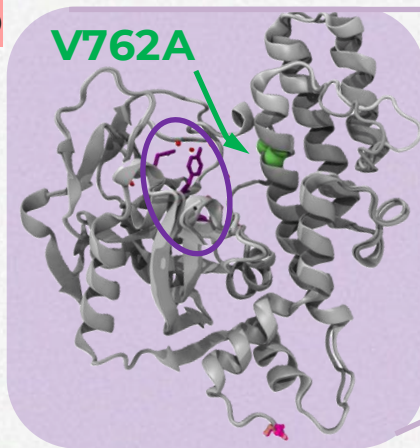
Evaluate: *Biological Relevance*



- Biological assembly (BA) = asymmetric unit? **No**
 - Correct way to model: **as a monomer**
 - Note: PDB file contains full asymmetric unit
- Mutations or engineered constructs? **Yes (V762A)**
 - `grep MUTATION 7AAA.pdb`
 - If **Yes**, revert or keep?

Revert (may influence active-site dynamics)

Active site

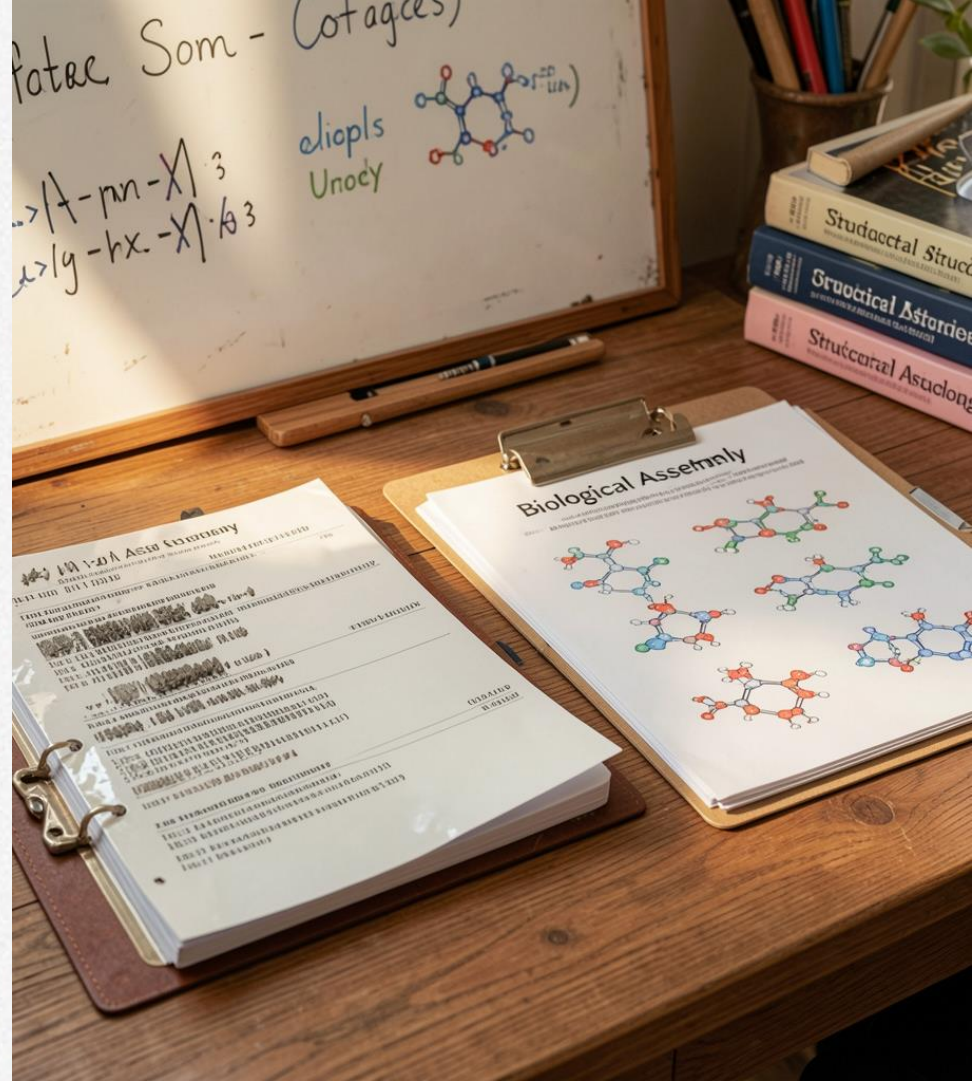


BA1 Chain A)

Asymmetric Unit (Chains A+B)

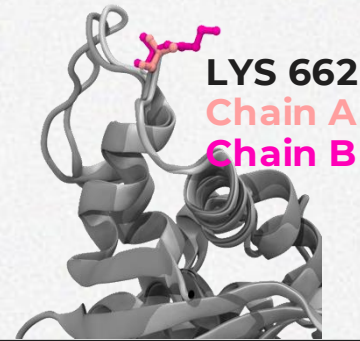
BA2 (Chain B)

Repair and Refine the Starting Protein Structure: *Structural* & *Chemical Refinement*



Repair and Refine: *Structural Refinement*

- Options:
 - Leave unresolved regions as-is (when appropriate)
 - Use an experimental template
 - Direct structural alignment
 - Build a model
 - Comparative (homology) modeling
 - Use AI (AlphaFold predictions):
 - AlphaFold DB
 - AlphaFold3



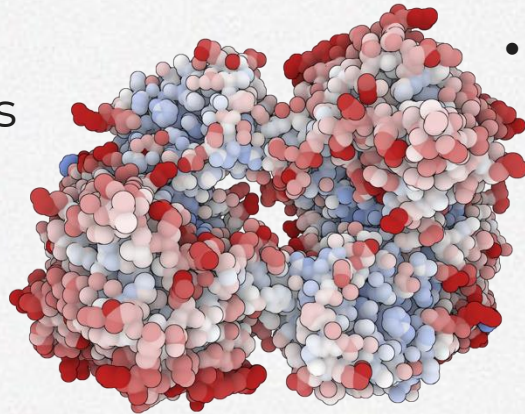
Modeller

Program for Comparative Protein
Structure Modelling by Satisfaction
Spatial Restraints



Repair and Refine: *Chemical Refinement*

- Assign protonation states
 - Physiological pH
 - Histidines
 - Asp/Glu
 - Lys/Arg
 - Termini
 - Catalytic residues
- Tools:
 - PROPKA
 - H++
 - PDB2PQR
- Verify disulfide bonds
 - Are they experimentally resolved?
 - Should any cysteines form disulfides?
 - `grep SSBOND 7AAA.pdb`



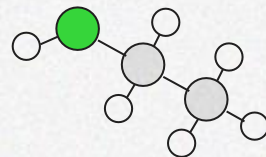


Apply the Workflow: *Decision Exercise*

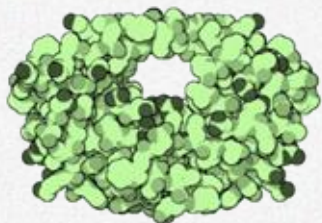
MD Starting Structure Checklist (high-level version):

- ☐ Found an appropriate structure
- ☐ Evaluated quality
- ☐ Checked completeness
- ☐ Checked biological relevance
- ☐ Identified any required repairs or refinements
- ☐ Assigned protonation states and verified disulfide bonds

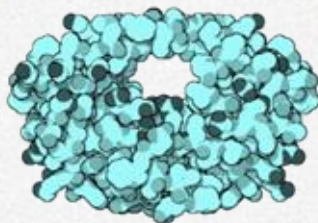
Apply: *Decision Exercise*



- **Prompt:** “You have just joined a lab. Your PI wants to simulate apo HIV protease. They send you three possible starting structures and ask which one you recommend.”
- **Goal:** Recommend one starting structure and justify your recommendation based on the features shown on the next slide.
Note: there is no perfect structure. Every structure involves tradeoffs.
- **Discuss:** First among your table-mates, then as a larger group!



Structure A: **1HVR**



Structure B: **3PHV**



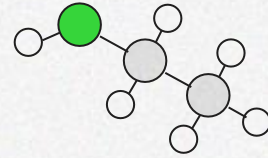
Structure C: **3BVB**

<u>Feature</u>	<u>1HVR</u>	<u>3PHV</u>	<u>3BVB</u>
Resolution	1.80 Å	 2.70 Å	1.30 Å 
Apo/Holo	Holo	 Apo	Holo
Alternate conformations	None	None	250 atoms! 
Waters	0	0	196 
Mutations	 Sequence Conflict (A95 vs UniProt C151)	None	Multiple engineered mutations 

Discuss:

- Which structure would you recommend and why?
- Is biological state more important than resolution?
- Would you remove a bound ligand, or start from an apo structure, if it meant accepting a lower-resolution structure?
- How can a high-resolution structure still contain uncertainty (many alt. confs.)?
- Would you choose the same structure if your PI wanted to simulate inhibitor binding instead of apo dynamics? Why or why not?

Key Session 1 Takeaways



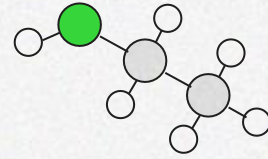
- Don't trust the download button!
 - Never assume the downloaded PDB is the correct starting structure.
- The experimentally resolved structure is NOT your simulation.
 - Experimental structure → Simulation requires many decisions.
 - Downloading a PDB is the beginning of system setup—not the end.
- Always ask *why* this structure was solved.
 - A PDB is not just coordinates. It's an experiment.
- High resolution does not mean the whole structure is high confidence!
 - The structure is only as good as the region you care about.
- Missing residues require modeling, missing atoms require repair.
- More chains in a PDB do not necessarily imply a biological oligomer.



Session 2

Preparing Ligands and Building Protein–Ligand Complexes

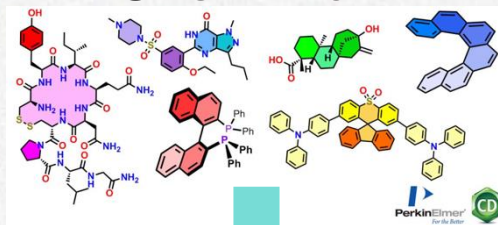
Learning Objectives



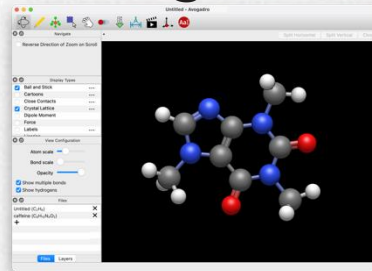
- **Find** or construct a suitable ligand structure (*Where do I get my ligand?*)
- Decide how to **build** the initial protein–ligand complex (*How do I get the ligand bound to my protein?*)
- **Evaluate** different strategies for generating binding poses (*When should I use a crystal structure, docking, or AI?*)
- **Apply** this workflow to a real protein–ligand system

Find a Starting Ligand Structure

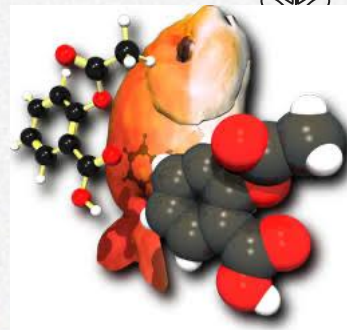
ChemDraw



Avogadro



Open-Source Cheminformatics
and Machine Learning



Other useful
tools: **RDKit**,
Open Babel,
CORINA

PubChem

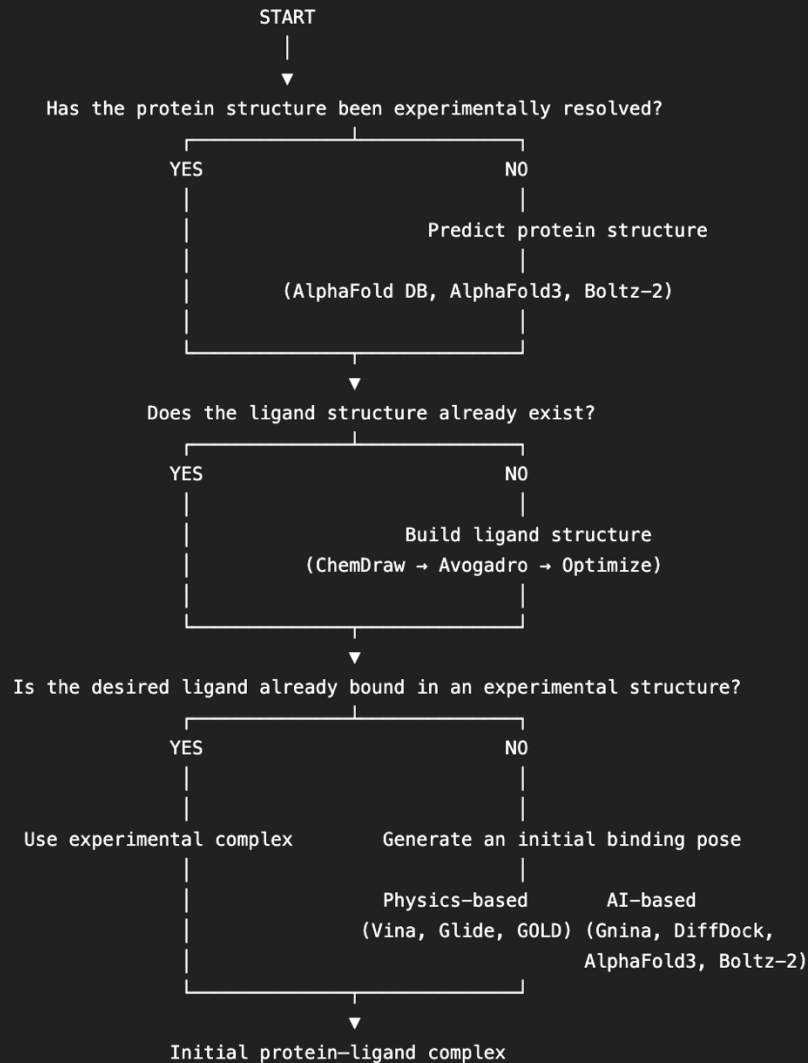


Curated Databases:
ChEMBL,
DrugBank, **ZINC**

Use **PDB ligand**,
PDBChem, or
PubChem for
experimental
coordinates

If ligand is unavailable,
use a **custom workflow**

Build the Initial Protein-Ligand Complex: *Decision Trees*



Build the Complex: *Decision Trees*

I want to model **Ligand X** bound to **Protein Y**

START

Has the protein structure been experimentally resolved?

YES

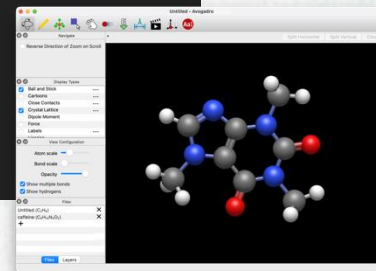
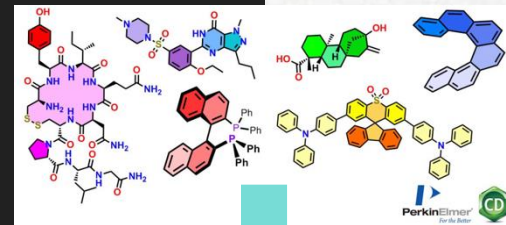
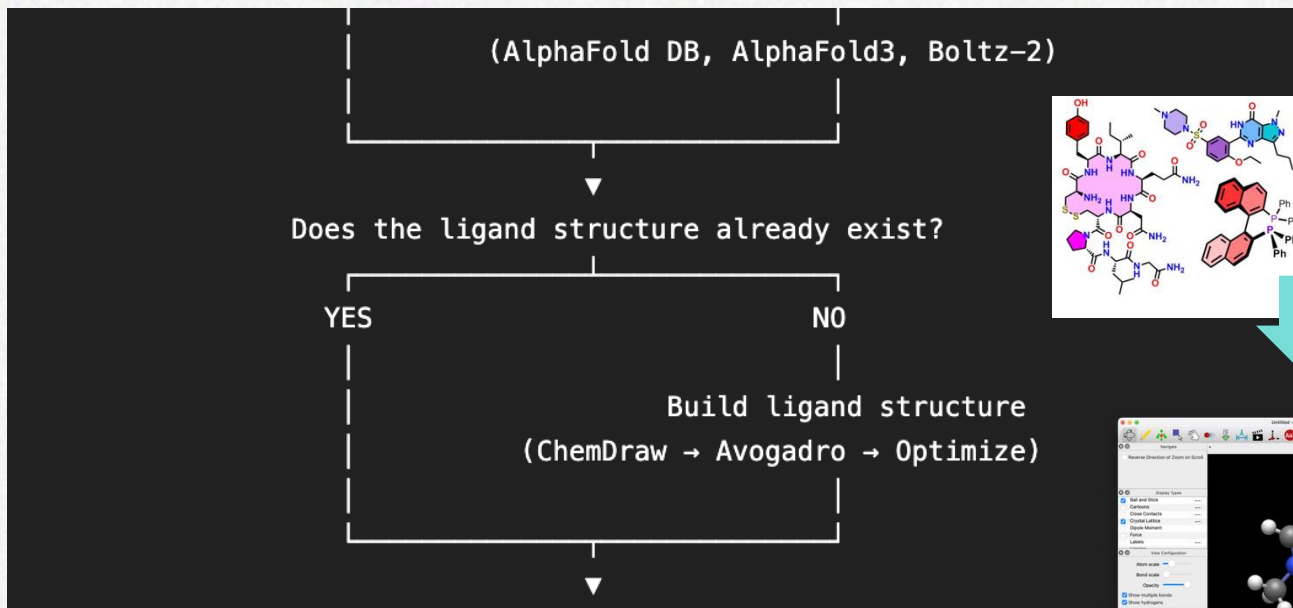
NO

Predict protein structure

(AlphaFold DB, AlphaFold3, Boltz-2)

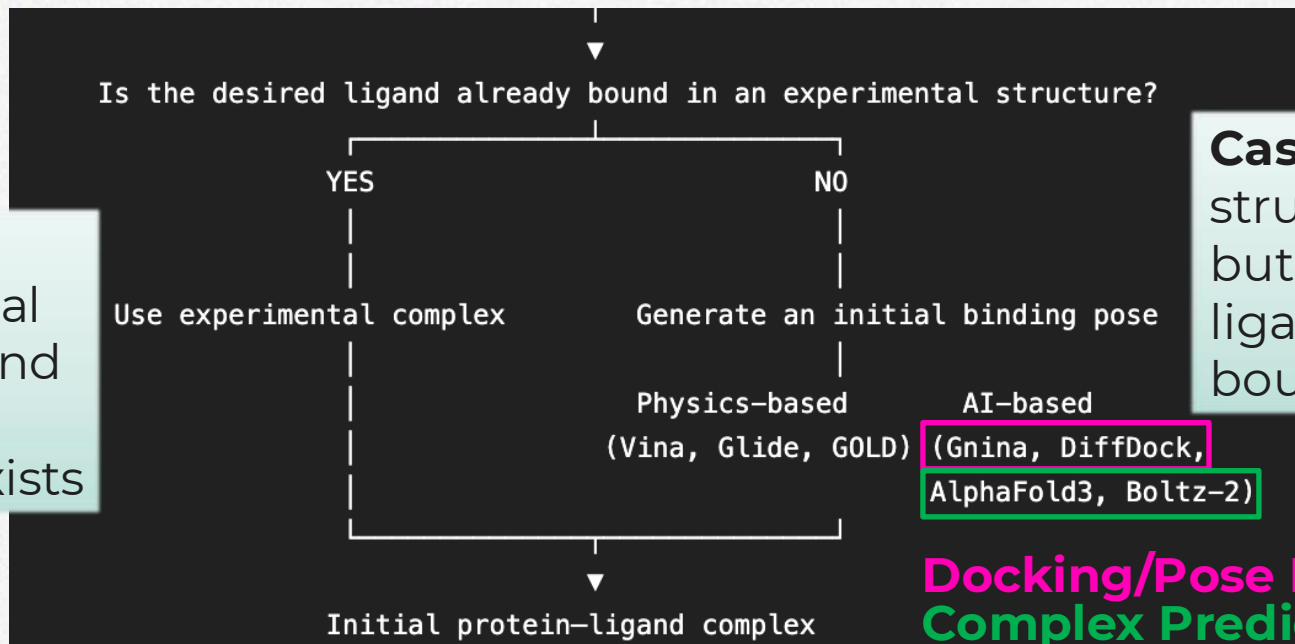
Case 1: No experimental protein structure exists

Build the Complex: *Decision Trees*



Build the Complex: *Decision Trees*

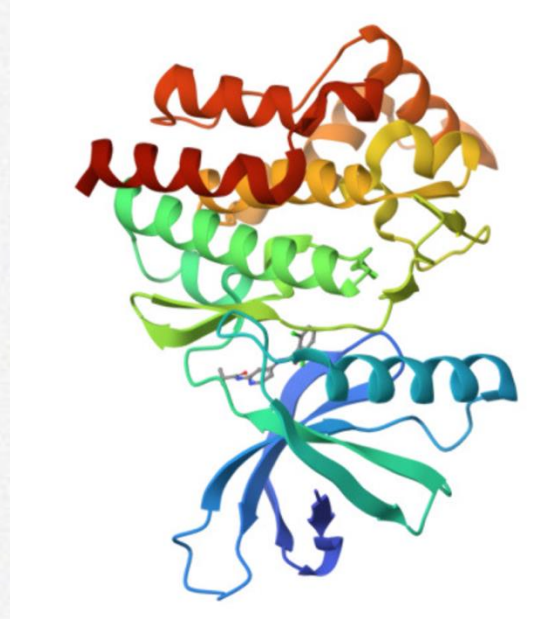
Case 2:
Experimental
protein-ligand
complex
structure exists



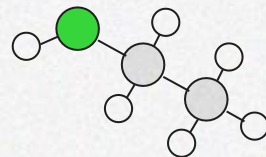
Case 3: protein
structure exists
but desired
ligand not
bound

Docking/Pose Prediction
Complex Prediction

Apply the Workflow: *Hands-On Exercise*



Apply: Part 1



Complete the MD Starting Structure Checklist (detailed version!):

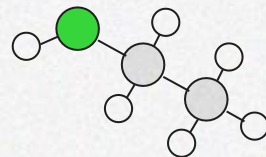
Evaluate: Structural Quality

- Highest resolution?
- X-ray, cryoEM, or NMR?
- Alternate residue conformations?
 - `awk '($1=="ATOM" || $1=="HETATM") && substr($0,17,1)!=" " {print}' 4GIH.pdb`
- Structural confidence?

Evaluate: *Biological Relevance*

- Truncated domains?
 - If **Yes**, do we need to cap the termini?
 - `grep '^ATOM' 4GIH.pdb | head -1`
 - `grep '^ATOM' 4GIH.pdb | tail -1`
- Fusion or purification tags?
 - `grep SEQADV 4GIH.pdb`
- Apo or holo?
- Bound ligands?
- Species?

Apply: Part 1

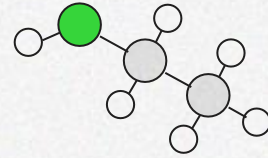


Complete the MD Starting Structure Checklist (detailed version!):

Evaluate: Biological Relevance

- Biological assembly (BA) = asymmetric unit?
 - Correct way to model?
- Mutations or engineered constructs?
 - `grep MUTATION 4GIH.pdb`
 - If **Yes**, revert or keep?

Apply: Part 1 – Solutions

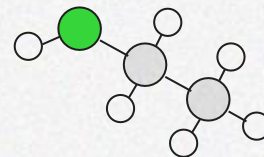


Complete the MD Starting Structure Checklist (detailed version!):

Evaluate: Structural Quality

- Highest resolution? **1.90 Å**
- X-ray, cryoEM, or NMR? **X-ray**
- Alternate residue conformations? **Yes**
 - 1 residue (**GLN A1102**) with alternate conformers: A and B
 - Choose **conformer B** due to higher occupancy (A=0.39; B=0.61)
 - **What did Schrödinger decide?!**
 - Solution file (protein.pdb) also has **conformer B!**
- Structural confidence? **High**

Apply: Part 1 – Solutions

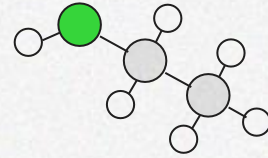


Complete the MD Starting Structure Checklist (detailed version!):

Evaluate: Structural Completeness

- Missing domains or residues? **Yes**
 - The deposited construct has **302** residues, but only **288** are modeled.
 - If **Yes**, do these need to be modeled in? **No (only at flexible termini)**
 - **What did Schrödinger decide?!**
 - Did not model in missing residues in flexible terminal regions
- Missing atoms? **No**
- Which (if any) crystallographic waters should be retained? **None**
 - **What did Schrödinger decide?!**
 - Did not retain crystallographic waters
- Missing cofactors or biologically relevant ions? **No**

Apply: Part 1 – Solutions

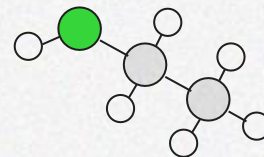


Complete the MD Starting Structure Checklist (detailed version!):

Evaluate: Biological Relevance

- Truncated domains? **Yes (not full-length TYK2; isolated JH1 domain)**
 - Evidence:
 - PDB page has “JH1” in title in parentheses
 - Sequence length on PDB page is 302, yet is 1187 on Uniprot page
 - PDB spans residues 890-1177, numbering from full-length TYK2
 - If **Yes**, do we need to cap the termini? **Yes**
 - **What did Schrödinger decide?!**
 - Solution file (protein.pdb) **caps with ACE and NME!**
- Fusion or purification tags? **No (none present in final modeled structure)**
- Apo or holo? **Holo (kinase crystallized with a bound inhibitor)**

Apply: Part 1 – Solutions

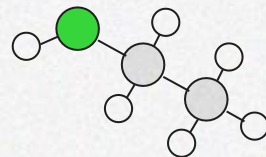


Complete the MD Starting Structure Checklist (detailed version!):

Evaluate: Biological Relevance

- Mutations or engineered constructs? **Yes (6)**
 - If Yes, revert or keep? **Keep**
 - **What did Schrödinger decide?!**
 - Not to revert mutations!

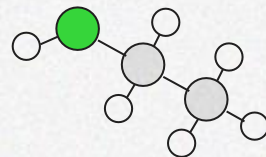
Apply: *Part 2*



Discuss (with tablemates and then as a larger group): using our decision tree, decide how best to obtain an initial complex with EJM_31 bound instead of EJM_46.

Options: Classical docking, AI-enhanced docking/pose prediction, AI complex prediction

Apply: Part 2 – Solutions



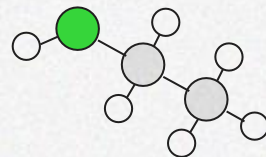
Discuss (with tablemates and then as a larger group): using our decision tree, decide how best to obtain an initial complex with EJM_31 bound instead of EJM_46.

Options: Classical docking, AI-enhanced docking/pose prediction, AI complex prediction

Complex pre-generated ahead of today's workshop using Schrödinger Glide, to ensure all have solid starting files for the upcoming analysis and free energy calculation sessions.



Key Session 2 Takeaways



- The crystallographic ligand IS NOT necessarily the ligand you want.
- Your biological question determines which ligand (or state) you should simulate.
- Docking and AI generate hypotheses—not answers.
- Different problems require different tools.
 - AlphaFold doesn't solve everything!
- The best starting model is the one that's most appropriate for the scientific question you're trying to answer.
- Experimental structures and predictions are complementary tools, each with strengths and limitations.